Time series: basic named processes

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The homework is provided on the last slide

Simulation of a stochastic process

To specify a particular stochastic process,
we have to define
the n-dimensional cumulative distribution function

$$F_{t_1,\ldots,t_n}(u_1,\ldots,u_n)$$

for the vector of random variables $(x_{t_1},\ldots,x_{t_n})$
for all n and all $t_1,\ldots,t_n$.
To simulate a realization (i.e. a time series)
from a given stochastic process, we can use

- $F_{t_1,\ldots,t_n}(u_1,\ldots,u_n)$, or
- specific modeling algorithms.

Simulation of a Gaussian process

Simulate a Gaussian process at points $0, 0.5, 1, 2, 3, 5, 7, 11$ with the mean function $m(t) = 1.2 + 0.14t + 0.9\sin(0.8t)$ and the covariance function $R(t, s) = 1.6e^{-0.7|t-s|}$.

```
® tp = c(0, 0.5, 1, 2, 3, 5, 7, 11)
mt = 1.2+0.14*tp+0.9*sin(0.8*tp)
n = length(tp)
CovMtp = matrix(0, nrow=n, ncol=n)
for(i in c(1:n))
{ for(j in c(1:n))
{ CovMtp[i,j] = 1.6*exp(-0.7*abs(tp[i]-tp[j]))
}}
GausP = mt+t(chol(CovMtp))%*%rnorm(n, 0, 1)
plot(tp, GausP, type="l", main="Gaussian process")
lines(tp, mt, col="red")
```

White noise process

A stochastic process ϵ_t, $t \in \mathbb{Z}$ is called a **white noise process** if the sequence $...\epsilon_{t-1}, \epsilon_t, \epsilon_{t+1}, \epsilon_{t+2}...$ consists of independent and identically distributed (iid) random variables with

$$\mathbf{E}\epsilon_t = m \text{ and } \mathrm{Var}(\epsilon_t) = \sigma^2.$$

From definition, we have $\mathrm{Cov}(\epsilon_t, \epsilon_s) = \begin{cases} \sigma^2, & t{=}s \\ 0, & t{\neq}s \end{cases}$.

A white noise process is called a **Gaussian white noise** if ϵ_t has the Gaussian (normal) distribution.

```
® n=100; m=3; sigma=1.2;
eps = rnorm(n, m, sigma)
plot(c(1:n), eps, type="p", pch=19, main="White noise")
```

See more `http://en.wikipedia.org/wiki/Colors_of_noise`

Random walk

A process x_t, $t \in \{1, 2, \ldots\}$ is called a **random walk process** if it is defined by $x_1 = \epsilon_1$ and

$$x_t = \epsilon_1 + \epsilon_2 + \ldots + \epsilon_t, \ t = 2, 3, \ldots,$$

where ϵ_t is a white noise process with mean 0 and variance σ^2.

Note that $x_t = x_{t-1} + \epsilon_t$ for $t = 2, 3, \ldots$.

Lemma. We have $\mathbf{E}x_t = 0$, $\mathrm{Var}(x_t) = t\sigma^2$, $\rho(t, s) \overset{t \leqslant s}{=} \sqrt{t/s}$, $\mathrm{Cov}(x_t, x_s) \overset{t \leqslant s}{=} t\sigma^2$.

```
® n=100; m=0; sigma=1.2; eps=rnorm(n, m, sigma);
x=rep(0,n); x[1]=eps[1];
for(t in (2:n)) x[t]=x[t-1]+eps[t];
plot(c(1:n), x, type="p", pch=19, main="Random walk")
```

Proof. Characteristics of the random walk

$$\mathbf{E}x_t = \mathbf{E}(\epsilon_1 + \epsilon_2 + \ldots + \epsilon_t) = \mathbf{E}\epsilon_1 + \mathbf{E}\epsilon_2 + \ldots + \mathbf{E}\epsilon_t = 0,$$

$$\mathrm{Cov}(x_t, x_s) = \mathbf{E}(\epsilon_1 + \epsilon_2 + \ldots + \epsilon_t)(\epsilon_1 + \epsilon_2 + \ldots + \epsilon_s) =$$

$$= \sum_{i=1}^{t} \mathbf{E}\epsilon_i^2 + \sum_{i\in\{1,\ldots,t\}} \sum_{j\in\{1,\ldots,s\}\setminus i} \mathbf{E}\epsilon_i\epsilon_j$$

$$= \sum_{i=1}^{t} \mathrm{Cov}(\epsilon_i, \epsilon_i) + \sum_{i\in\{1,\ldots,t\}} \sum_{j\in\{1,\ldots,s\}\setminus i} \mathrm{Cov}(\epsilon_i, \epsilon_j)$$

$$= \sum_{i=1}^{t} \mathrm{Var}(\epsilon_i) = \sum_{i=1}^{t} \sigma^2 = t\sigma^2,$$

$$\rho(t,s) = \frac{\mathrm{Cov}(x_t, x_s)}{\sqrt{\mathrm{Cov}(x_t, x_t)\mathrm{Cov}(x_s, x_s)}} = \frac{t\sigma^2}{\sqrt{t\sigma^2 s\sigma^2}} = \sqrt{t/s}.$$

Brownian motion / Wiener process

A process W_t, $t \in [0, \infty)$ is called a **Brownian motion** (or **Wiener process**) if W_t has continuous realizations and

- $W_0 = 0$,
- increments follow a normal distribution, $W_s - W_t \sim N(0, s-t)$ for all $0 \le t \le s$,
- increments $W_s - W_t$ and $W_v - W_u$ are independent for all $0 \le t < s \le u < v$.

Lemma. $\mathbf{E}W_t = 0$ and $R(t,s) = \mathrm{Cov}(W_t, W_s) = \min(t,s)$ for all $t, s \ge 0$, or, equivalently, $R(t,s) \overset{t \leqslant s}{=} t$. A Brownian motion sampled on an equidistant grid is equivalent to a random walk.

Ⓡ *n=100; tm=seq(0, 9, length.out=n);*
eps=rnorm(n,0,sqrt(tm[2]-tm[1]));
W=rep(0,n); W[1]=0; for(t in (2:n)) W[t]=W[t-1]+eps[t];
plot(tm, W, type="l", ylim=c(-2,2), main="Wiener process")

Brownian bridge

A process B_t, $t \in [0,1]$ is called a **Brownian bridge** if

- $B_1 = 0$,
- B_t is a Brownian motion on the interval $[0,1]$.

Note that $\mathbf{E}B_t = 0$ and $R(t,s) = \mathrm{Cov}(B_t, B_s) \overset{t \leqslant s}{=} t(1-s)$.
The process B_t can be written as $B_t = W_t - tW_1$.

```
® n=100; tm=seq(0, 1, length.out=n);
eps=rnorm(n, 0, sqrt(tm[2]-tm[1]))
W=rep(0, n); W[1]=0;
for(t in (2:n)) W[t]=W[t-1]+eps[t];
B=W-tm*W[n];
plot(tm, B, type="l", ylim=c(-.2,.2), main="Brownian bridge")
mtext(paste("n=", n, sep=""), 3)
```

Poisson process

A process P_t, $t \in [0, +\infty)$ is called a **Poisson process** if

- $P_0 = 0$,
- The distribution of P_t is a Poisson distribution $\mathcal{P}(\lambda t)$.
- Independent increments, $P_s - P_t$ and $P_v - P_u$ are independent for all $0 \leq t < s \leq u < v$.
- Stationary increments, $\mathbb{P}(P_s - P_t < u) = \mathbb{P}(P_{s-t} - P_0 < u)$ for all $0 \leq t < s$ and u.
- $P_s - P_t \leq 1$ if $s - t$ is small enough, $0 \leq t < s$.

Interpretation of Poisson process

The Poisson process can be considered as a **counting process** because P_t is integer for all t.
Conditions are interpreted as follows

- Independent increments: the numbers of occurrences counted in disjoint intervals are independent of each other.
- Stationary increments: the probability distribution of the number of occurrences counted in any time interval only depends on the length of the interval.
- $P_s - P_t \leq 1$ if $s - t$ is small enough: no counted occurrences are simultaneous.

Poisson process as renewal process

The Poisson process can be considered as a **jump process** because P_t has jumps of size 1 and the time between consecutive jumps has an exponential distribution.

```
® n=30; a=1.6; tbetween=rexp(n, a); P=c(1:n);
tjumps=rep(0,n); tjumps[1]=tbetween[1];
for(i in (2:n)) tjumps[i]=tjumps[i-1]+tbetween[i];
plot(c(0,tjumps[1]),c(0,0),type="l",main="Poisson process",
xlim=c(0,1.1*n/a), ylim=c(0,n), xlab="time", ylab="P");
for(i in (2:n))
lines(c(tjumps[i-1],tjumps[i]),c(P[i-1],P[i-1]),type="l")
mtext(paste("n=", n, " a=", a, sep=""),3)
```

Moving average

A stochastic process y_t, $t \in \mathbb{Z}$ is called a **moving average** process of the process x_t if y_t has the form

$$y_t = \sum_{i=1}^{n} c_i x_{t+1-i}$$

for some n and $c_1, \ldots, c_n$.

If $c_i \geq 0$ and $\sum_{i=1}^{n} c_i = 1$, then y_t is a **smoothing** of the process x_t.

The popular choice of c_i is $c_i = 1/n$.

Moving average of white noise

Task. Let $y_t = (\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})/3$, where ϵ_t is a white noise process with mean 0 and variance σ^2. Compute the mean function $\mathbf{E}y_t$ and the covariance function $R(t,s) = \mathrm{Cov}(y_t, y_s)$.

$$\mathbf{E}y_t = \mathbf{E}(\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})/3 = (\mathbf{E}\epsilon_t + \mathbf{E}\epsilon_{t-1} + \mathbf{E}\epsilon_{t-2})/3 = 0.$$

Let us now compute $R(t,t)$, $R(t,t+1)$, $R(t,t+2)$, $R(t,t+3), \ldots$.
Since $\mathbf{E}(\epsilon_t\epsilon_s) = \mathrm{Cov}(\epsilon_t, \epsilon_s) = 0$ for $t \neq s$, we have

$$R(t,t) = \mathbf{E}(\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})(\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})/9 =$$
$$= [\mathbf{E}(\epsilon_t\epsilon_t) + \mathbf{E}(\epsilon_{t-1}\epsilon_{t-1}) + \mathbf{E}(\epsilon_{t-2}\epsilon_{t-2})]/9$$
$$= [\mathrm{Var}(\epsilon_t) + \mathrm{Var}(\epsilon_{t-1}) + \mathrm{Var}(\epsilon_{t-2})]/9 = \sigma^2/3.$$

Moving average of white noise, 2

Then

$$R(t,t+1) = \mathbf{E}(\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})(\epsilon_{t+1} + \epsilon_t + \epsilon_{t-1})/9 =$$

$$= [\mathbf{E}(\epsilon_t\epsilon_t) + \mathbf{E}(\epsilon_{t-1}\epsilon_{t-1})]/9$$

$$= [\mathrm{Var}(\epsilon_t) + \mathrm{Var}(\epsilon_{t-1})]/9 = 2\sigma^2/9.$$

Also

$$R(t,t+2) = \mathbf{E}(\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})(\epsilon_{t+2} + \epsilon_{t+1} + \epsilon_t)/9 =$$

$$= [\mathbf{E}(\epsilon_t\epsilon_t)]/9 = [\mathrm{Var}(\epsilon_t)]/9 = \sigma^2/9.$$

Finally

$$R(t,t+3) = \mathbf{E}(\epsilon_t + \epsilon_{t-1} + \epsilon_{t-2})(\epsilon_{t+3} + \epsilon_{t+2} + \epsilon_{t+1})/9 = 0$$

and similarly

$$R(t,s) = 0 \text{ for } s \geq t+3.$$

Properties of the covariance function

Take some function $R(t)$.

Does a wide-sense stationary process with covariance function $R(t)$ exist?

Answer is NO. The function $R(t)$ should satisfy some properties in order to be a covariance function.

Positive semidefiniteness

Theorem. The covariance function $R(t)$ of a stationary process x_t satisfies

$$\sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j R(t_i - t_j) \geq 0$$

for all n, $t_1, \ldots, t_n$ and $c_1, \ldots, c_n$. This property is called **positive semi-definiteness**.

Proof. Consider the random variable $\xi = \sum_{i=1}^{n} c_i x_{t_i}$. Then

$$\mathrm{Var}(\xi) = \mathrm{Cov}(\xi, \xi) = \mathrm{Cov}\Big(\sum_{i=1}^{n} c_i x_{t_i}, \sum_{j=1}^{n} c_j x_{t_j}\Big)$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j \mathrm{Cov}(x_{t_i}, x_{t_j}) = \sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j R(t_i - t_j).$$

Since $\mathrm{Var}(\xi) \geq 0$, we obtain the required inequality.

Short-range and long-range dependence

A stationary process is said to have **short memory** (**short-range dependence**) if the series $\sum_{k=1}^{\infty} R(k)$ converges, i.e. this sum is finite.

A stationary process is said to have **long memory** (**long-range dependence**) if the series $\sum_{k=1}^{\infty} R(k)$ diverges, i.e. this sum is infinite.

Task. Consider a stationary process x_t with $R(t) = \frac{1}{(1+|t|^{\beta})^{\alpha/\beta}}$, $\alpha > 0, \beta > 0$. If $\alpha \in (0, 1]$, then x_t is a long memory stationary process because the series

$$\sum_{k=1}^{\infty} \frac{1}{(1+|k|^{\beta})^{\alpha/\beta}} \text{ has the same convergence as } \sum_{k=1}^{\infty} \frac{1}{k^{\alpha}},$$

which diverges when $\alpha \in (0, 1]$ and converges when $\alpha > 1$.

The spectral density function

Consider a stationary process x_t, $t \in \mathbb{Z}$, with the covariance function $R(k)$.

A function $f(\lambda)$ defined by

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} R(k) \cos(k\lambda)$$

is called the **spectral density function** (or **spectrum**).

There is a backward transformation:

$$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda) f(\lambda) d\lambda.$$

Note that $f(\lambda) = f(-\lambda)$, $f(\lambda) \geq 0$, $\lambda \in [-\pi, \pi]$.

Homework

Apply AI tools to deepen your understanding of basic names processes.

- **Conceptual review:** Use AI to explain the intuitive differences between a random walk, Brownian motion, and a Poisson process.
- **Comparative simulation:** Generate R code to plot 5 realizations of a random walk versus 5 realizations of a Brownian bridge on the same graph.
- **Parameter sensitivity:** Run simulations of a Poisson process with varying intensity λ and observe how it changes the frequency of jumps.
- **Implementation:** Compare R code generated by AI that uses specialized libraries (like `stats`) versus code written from scratch using loops.
- Solve exam-like exercises for L4.